Chapter 1

Latent Space Analysis

The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations emerge progressively through the hidden layers during training and play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how neural networks achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep learning introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, neural networks gradually construct representations that become increasingly relevant to the task under consideration. As discussed in Chapter **??**, these representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training.

The study of these representations has therefore become a central topic in deep learning research. In particular, researchers seek to understand how information is organized within these representations, what concepts they encode, and how they contribute to the model's behavior. The collection of representations produced by a hidden layer is referred as a latent space.

In this chapter, we first make explicit the working hypothesis that underlie most research on latent space analysis. To do so, we introduce the notions of abstraction, concept, latent space, and semantics. Once the notions of this conceptual framework has been established, we review the main methods that aim to identify and extract concepts from latent representations.

1.1 Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on latent space analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, latent space, concept, and semantics. It should therefore be understood as a set of shared intuitions that guide the study and interpretation of neural representations. The goal of this section is to make this conceptual framework explicit by formally defining its main components.

A central motivation for this hypothesis is the generalization ability of neural networks observed after training, as discussed in Section **??**. Formally, a model is said to generalize when, for an input $x \in X$ not seen during training, it still produces an output $\hat{y}$ close to the true output y. Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task.

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

Through this abstraction process, a neural network does not directly map x to y. Instead, as illustrated in Figure **??**, it computes a sequence of intermediate activations across hidden layers. These activations correspond to internal encodings of the input, which we refer to as representations in the sense of Definition 1.1.2.

Definition 1.1.2: Representation

A representation is the internal encoding of an input produced by a trained neural network as a result of abstraction. It corresponds to the activations of one or several layers and provides a transformed view of the input used by the model for prediction.

Each hidden layer refines the representation of the input. It progressively transforms it into a structure more directly aligned with the output space. This process can be described as a sequence of transformations:

$$x \to h^{(1)} \to h^{(2)} \to \cdots \to h^{(L-1)} \to y$$

where each $h^{(l)}$ is a representation in the sense of Definition 1.1.2.

The set of all such representations produced over a dataset forms what is commonly referred to as the latent space of the model.

Definition 1.1.3: Latent Space

The latent space is the set of all representations produced by a layer or model over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

The existence of meaningful structure in the latent space is closely related to generalization. If representations capture reusable regularities across inputs, then unseen inputs can be associated with previously learned structures.

These recurring structures are typically referred to as concepts, as defined in Definition 1.1.4.

Definition 1.1.4: Concept

A concept is a recurring activation pattern in the latent space that is shared across multiple inputs. Concepts are generally distributed and overlapping, such that a single representation typically involves multiple concepts.

Under this view, a representation can be interpreted as a composition of multiple concepts, each contributing to the model's prediction.

This raises the question of whether these learned concepts align with those used by humans to interpret the same data, as formalized in Definition 1.1.5.

Definition 1.1.5: Semantics

Semantics refers to the alignment between concepts in the latent space and concepts used by a human observer. A structure is semantic if it can be consistently associated with a human-interpretable concept.

When such alignment exists, we say that the learned concepts have semantic meaning. However, neural networks may also develop highly effective internal concepts that remain difficult to interpret from a human perspective.

It is important to emphasize that this conceptual framework remains largely hypothetical. Current mathematical tools do not provide a complete characterization of deep neural network internals, nor a formal proof of the existence of concepts in learned representations. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and latent space analysis. As a result, many research efforts focus on analyzing latent spaces, identifying recurring patterns in activations, and understanding how such structures emerge during training.

The remainder of this chapter is devoted to reviewing methods for analyzing latent spaces and extracting the concepts they encode.

1.2 Probing

Probing methods aim to determine whether a given human-defined concept is encoded within the internal representations of a neural network. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, it is likely that the network encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of interpretable concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concept. The data are subsequently projected into the latent space of the pretrained model. From these latent representations, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe, since its role is to determine whether a concept is present or not. The role of this probe is therefore to assess whether the information associated with the concept is accessible from the network's internal representations.

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to a representation explicitly encoded in the latent space. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the neural network and potentially usable during decision making.

If the classifier achieves performance significantly above chance level, this suggests that the latent space contains exploitable information related to the considered concept. However, the fact that a concept can be decoded from latent representations does not necessarily imply that the model actually uses this information to perform its task.

To ensure that representations play a causal role in the model's output, it is possible to perform interventions on the latent representations. An intervention consists of modifying a latent representation in a controlled manner by activating or deactivating a given concept, and then studying its impact on the model's output. If the resulting change in the model's output is consistent with the applied perturbation, this suggests that the representation had a causal role.

The main limitation of this approach is that the concepts under investigation must be defined in advance before any probing analysis can be performed. This introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in latent space are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network's output under a given perturbation.

This set of limitations makes probing a highly human-centric approach, strongly dependent on the choice of investigated concepts and on the interpretation of interventions. To overcome this bias, other methods for studying latent spaces have been developed, particularly those aiming to discover concepts in an unsupervised manner.